

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Currency Conversion Agent | API-Workflow Domain (Finance)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external API integration

1. Problem Statement

Finance and procurement teams frequently receive emails asking for a currency conversion — for example, converting a vendor quote from USD to INR before approving a purchase order, or checking today's exchange rate before releasing an international payment. Today, staff manually look up the rate on a banking portal or search engine, calculate the converted amount, and reply by email — a small but repetitive task that interrupts higher-value work and can introduce manual calculation errors.

Design and build an Autonomous UiPath Agent, the AI Currency Conversion Agent, that reads an incoming email requesting a currency conversion, extracts the amount and currency pair, calls a live Currency Exchange Rate API workflow to fetch the current rate, computes the converted amount, and independently replies with the result — without any human approval step.

2. System Prompt

You are an AI Currency Conversion Agent for the Finance department.
Your goal is to analyze currency conversion request emails and provide an accurate, up-to-date converted amount using a live exchange rate API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Currency Conversion Request
 - Exchange Rate Enquiry (rate only, no amount to convert)
 - Other
3. If it is Currency Conversion Request or Exchange Rate Enquiry:
 - Extract the source currency, target currency, and amount (if provided)
 - Call the Currency Exchange Rate API workflow with the currency pair
 - Compute the converted amount using the returned rate
 - Generate a professional reply stating the rate, the converted amount, and the timestamp of the rate
4. If it is Other:
 - Skip processing

```
Rules:
- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the amount is not provided, only
  return the current exchange rate
- Do not invent exchange rate values – always use the API response
- Return output in structured format
```

3. User Prompt

```
Analyze the following currency conversion request email and provide the
converted amount using the current exchange rate:
{{conversion_email_text}}
```

4. Input Schema

```
{
  "conversion_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "source_currency": "string",
  "target_currency": "string",
  "amount": "string",
  "exchange_rate": "string",
  "converted_amount": "string",
  "rate_timestamp": "string",
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- source_currency (text) — e.g., "USD"
- target_currency (text) — e.g., "INR"
- amount (text) — numeric string, optional

Configure the workflow to call a live exchange-rate API endpoint (e.g., an HTTPS GET request to a currency-rates provider) and parse the JSON response for the rate and last-updated timestamp.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
source_currency	base_currency
target_currency	quote_currency
amount	amount_to_convert

8. Updated System Prompt — Tool Invocation Logic

```
If the email is classified as Currency Conversion Request or Exchange
Rate Enquiry:
- Call the get_exchange_rate API workflow with source_currency and
  target_currency.
- Parse the API response for 'rate' and 'last_updated'.
- If an amount was provided, compute converted_amount = amount * rate
  and include it in the reply.
- If no amount was provided, reply with only the current rate and
  timestamp.
- Use the generate_reply GenAI activity to draft a professional reply
  email with the rate, converted amount (if applicable), and
  timestamp – fully autonomously, with no manual approval required.

If the email is classified as Other:
- Take no further action.
```

9. Tools & Activities Summary

Tool / Activity	Type
get_exchange_rate	API Workflow (HTTP Request to a live currency exchange rate API)
generate_reply	GenAI Activity (LLM prompt-based text generation)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live API, computes the conversion, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and perform a calculation based on the result
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder